

Toward Scalable Bayesian Uncertainty for Machine Learning with the Ray Tracing Sampler

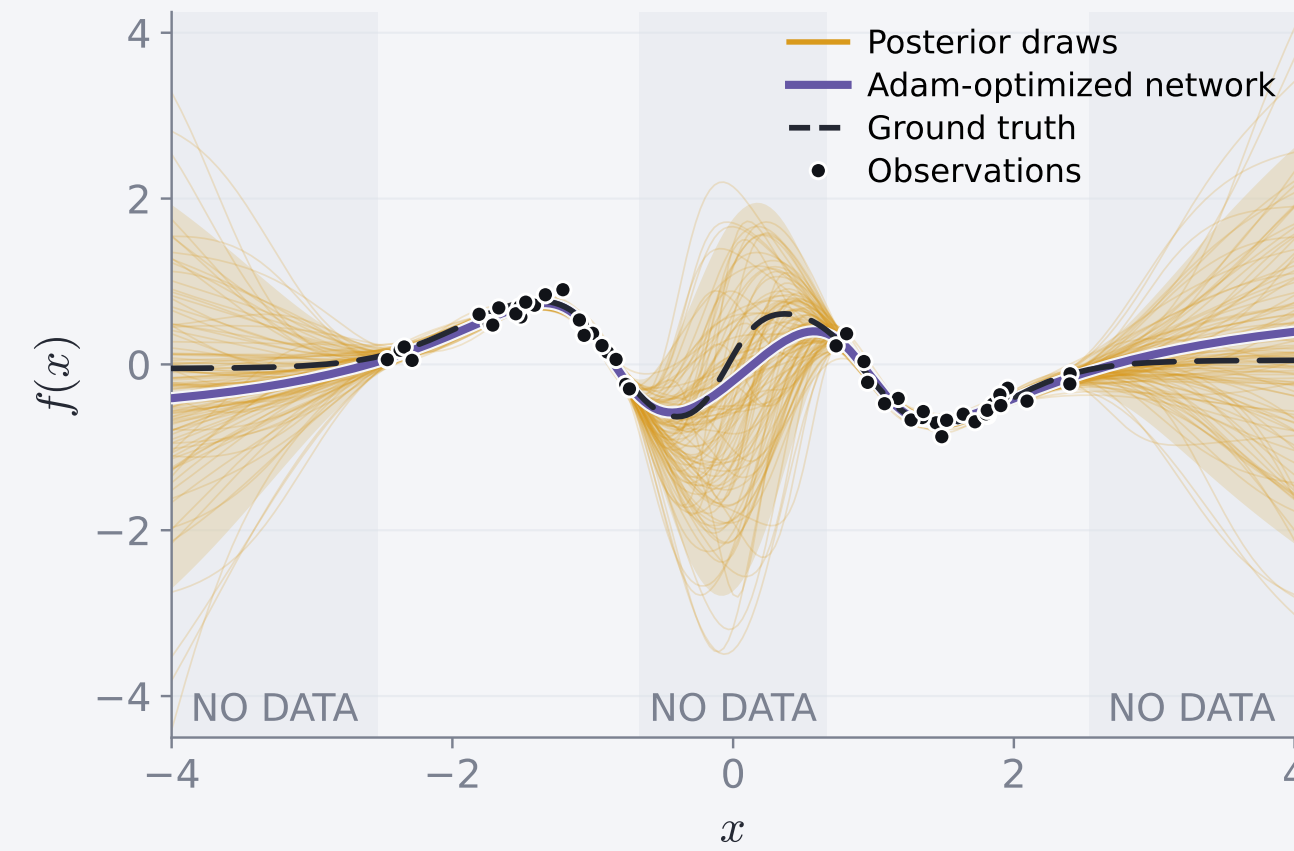
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Why sample a neural network?

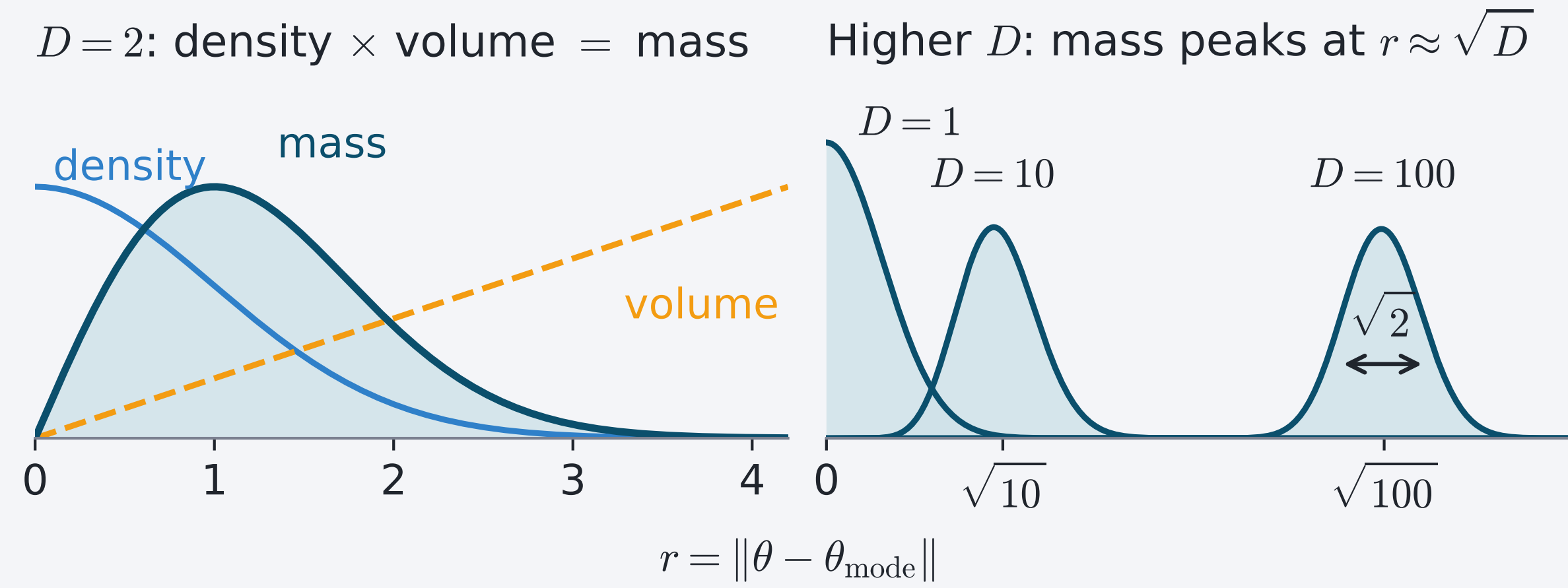


Ordinary training: one set of weights $\rightarrow$ one function.

Bayesian sampling: a distribution over weights $\theta \rightarrow$ uncertainty over functions.

$$\pi(\theta) = p(\theta | \mathcal{D}) \propto \underbrace{p(\mathcal{D} | \theta)}_{\text{fit}} \underbrace{p(\theta)}_{\text{prior}}$$

Where the probability lives



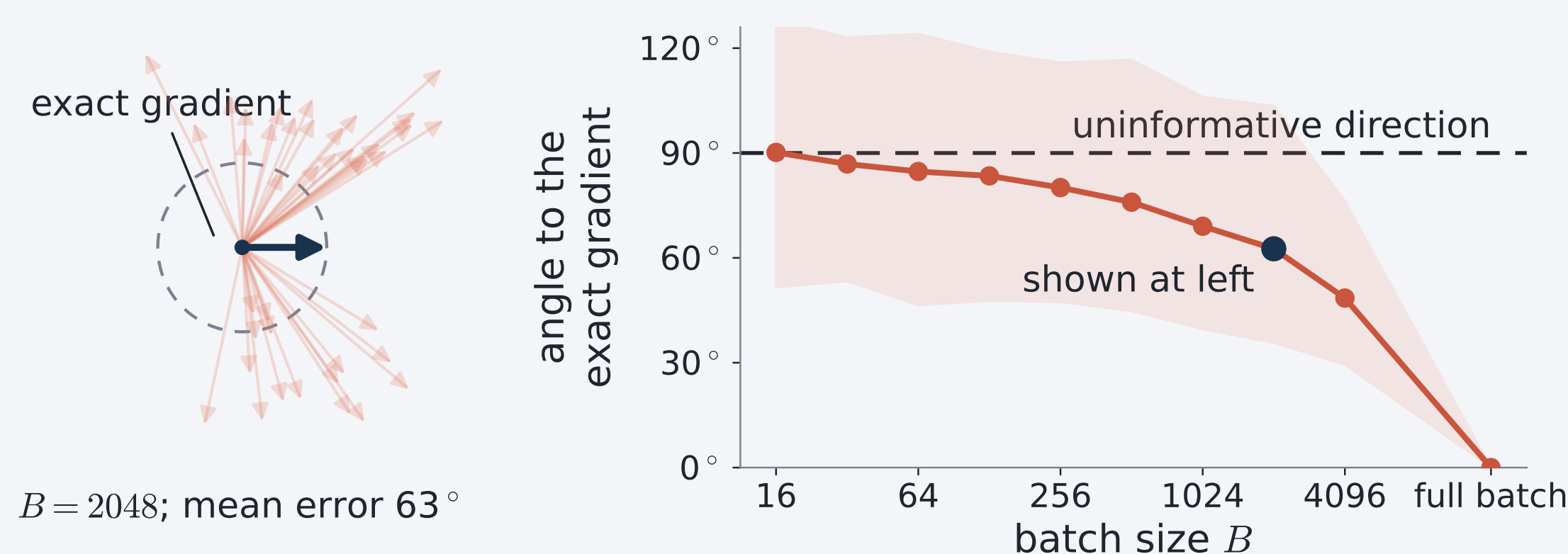
Because volume grows as r^{D-1} , probability concentrates in the *typical set*: a thin shell near $\|\theta\|^2 \approx D$. A correct sampler therefore spends almost all its time on this shell.

MCMC in the era of big data

Markov chain Monte Carlo (MCMC) generates a sequence of correlated samples targeting π . Gradient-based samplers use $\nabla \log \pi(\theta)$ to guide their updates. Computing this gradient exactly uses all N data points. A minibatch of size $B \ll N$ is unbiased and roughly N/B times cheaper, but never exact:

$$\hat{g}_B(\theta) = \nabla \log p(\theta) + \frac{N}{B} \sum_{i \in B} \nabla \log p(\mathcal{D}_i | \theta), \quad \text{Cov}[\hat{g}_B] \propto 1/B.$$

Example: UCI Superconductivity Bayesian linear regression ($D = 81$, $N = 17,010$).



$B = 2048$; mean error 63°

Code and experiments:

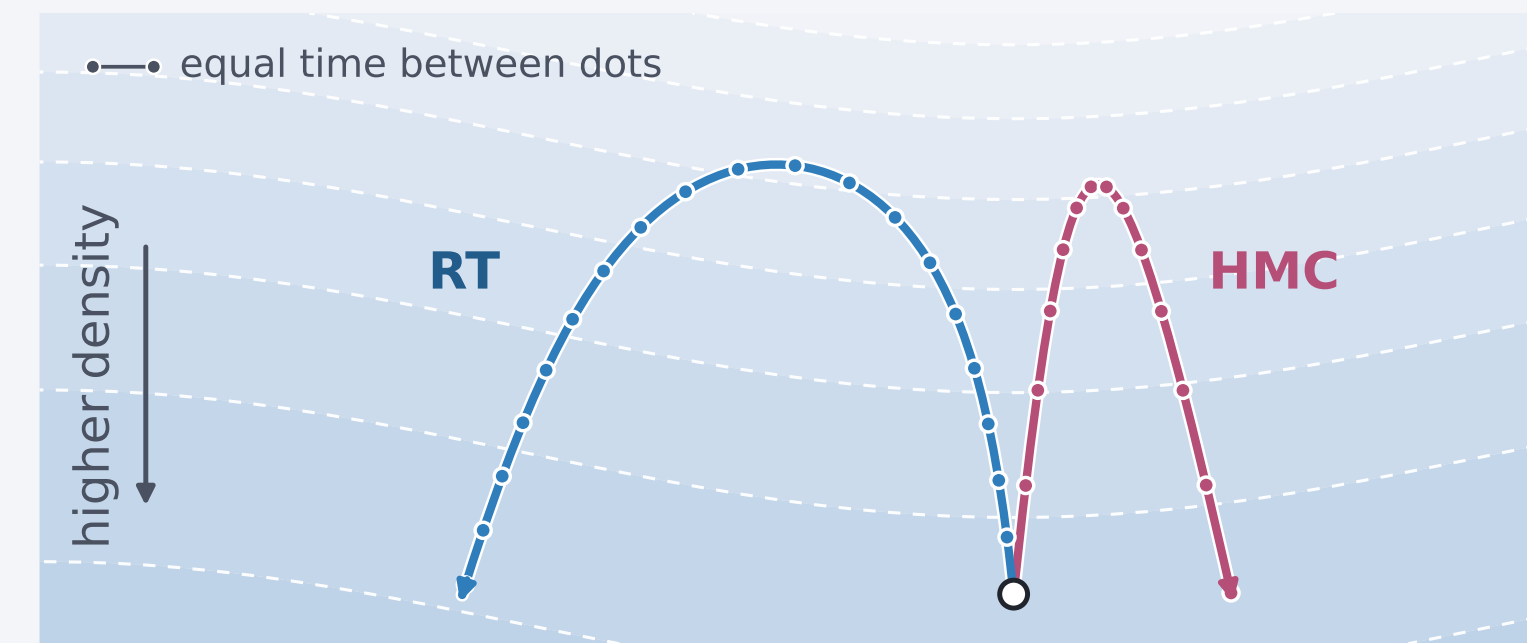
github.com/YasirAlsugair/

[ray-tracing-sampler-experiments](#)



HMC and ray tracing under gradient noise

Hamiltonian Monte Carlo (HMC)^[1] and ray tracing (RT)^[2] both use the log-posterior gradient to build trajectories. HMC treats it as a force, changing a particle's speed and direction. RT treats the posterior as a refractive medium; its ray moves at unit speed, so the gradient changes only direction.

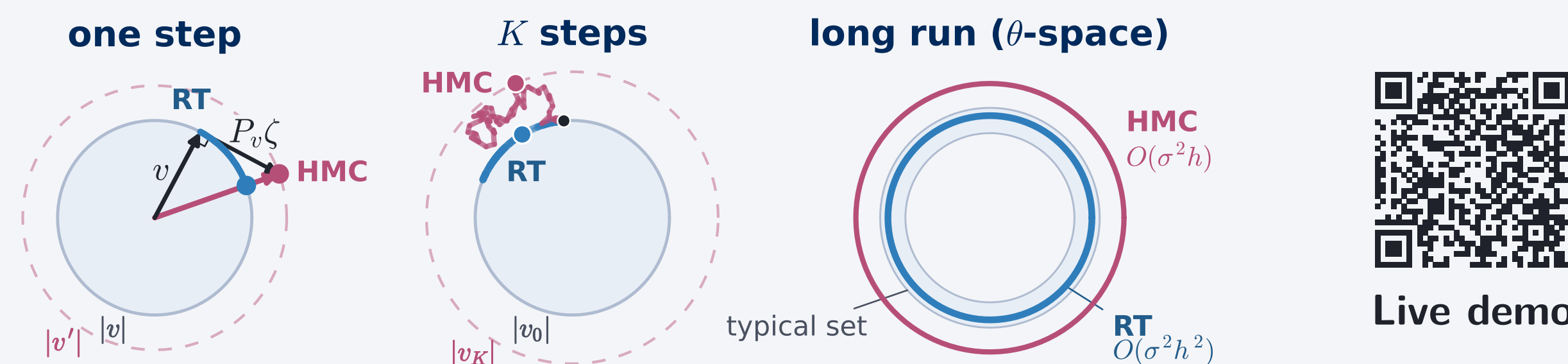


1. One noisy step

Without a Metropolis correction, mean-zero noise raises HMC's expected kinetic energy and inflates its speed. For RT, with isotropic error $\sigma\zeta$ and $n(\theta) = \pi(\theta)^{1/(D-1)}$, the unit direction obeys

$$\frac{du}{ds} = P_u \left(\nabla \log n(\theta) + \frac{\sigma}{D-1} \zeta \right), \quad P_u = I - uu^\top.$$

P_u projects onto the tangent plane, so noise changes u 's direction but not its norm.



2. Along a trajectory

With step size h and trajectory length S , the S/h noisy kicks accumulate as $\frac{S}{h} \cdot O(\sigma^2 h^2) = O(\sigma^2 h)$. Let $\sigma_c(h) \propto h^{-p}$ be the largest noise scale keeping long-run bias below a fixed tolerance. Unadjusted minibatch HMC has $p = \frac{1}{2}$.

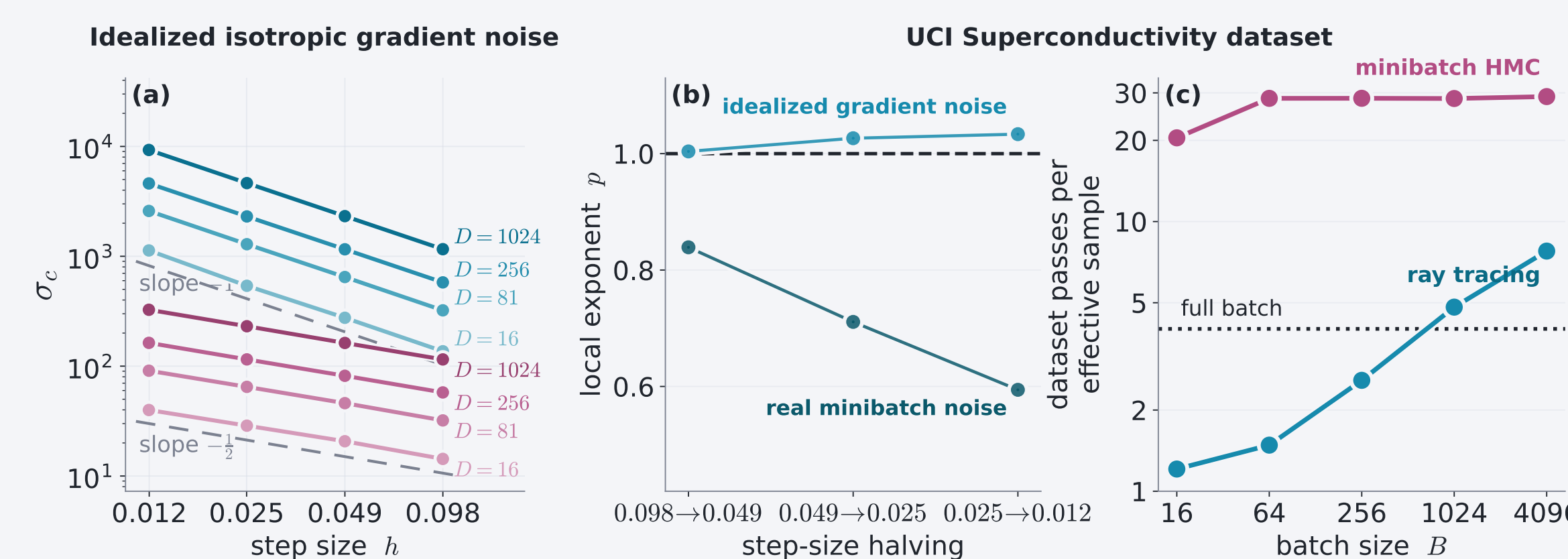
3. Long-run bias

For RT, isotropic rotations leave the uniform direction distribution unchanged, so the $O(\sigma^2 h)$ effect creates no leading bias:

$$b_{\text{RT}}(h, \sigma) = O(\sigma^2 h^2) + o(\sigma^2) \\ \Rightarrow \sigma_c(h) = \Omega(h^{-1}), \quad p \geq 1$$

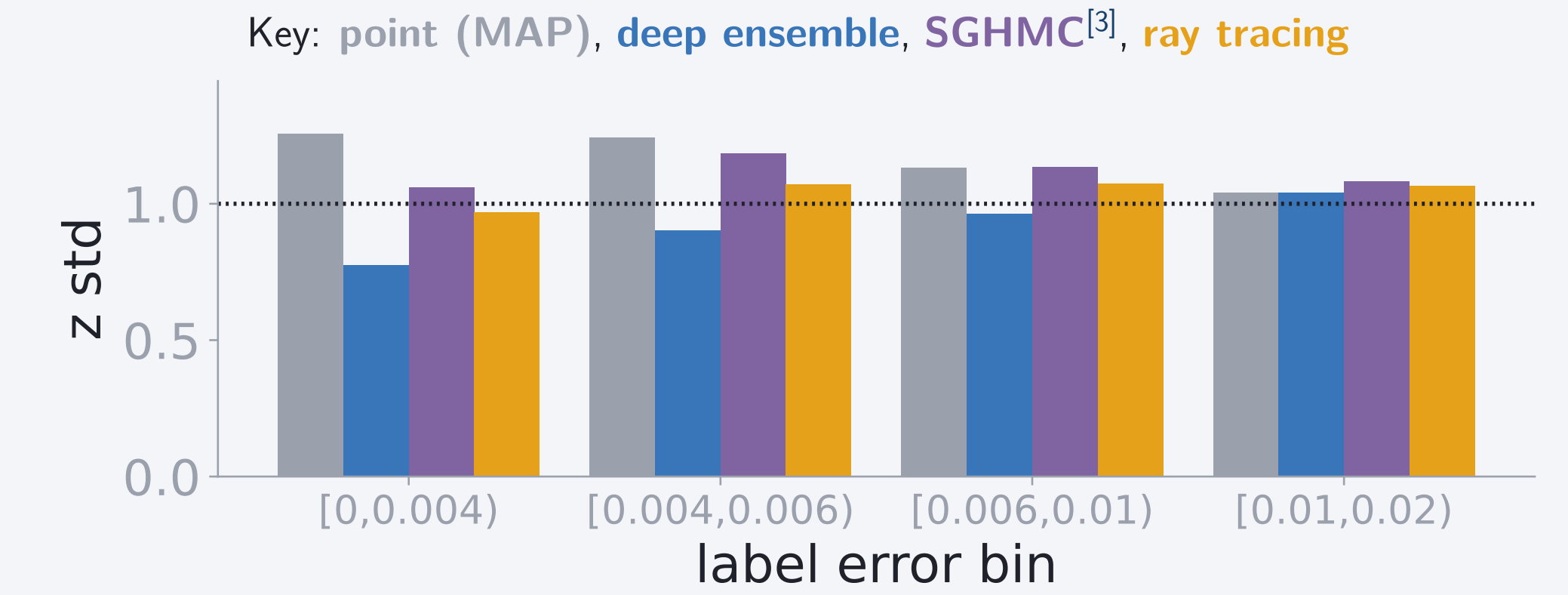
Measured noise tolerance and sampling cost

Across $D = 16$ – 1024 : $\sigma_c \propto h^{-1}$ (RT) versus $h^{-1/2}$ (HMC); real minibatch noise weakens RT's scaling.

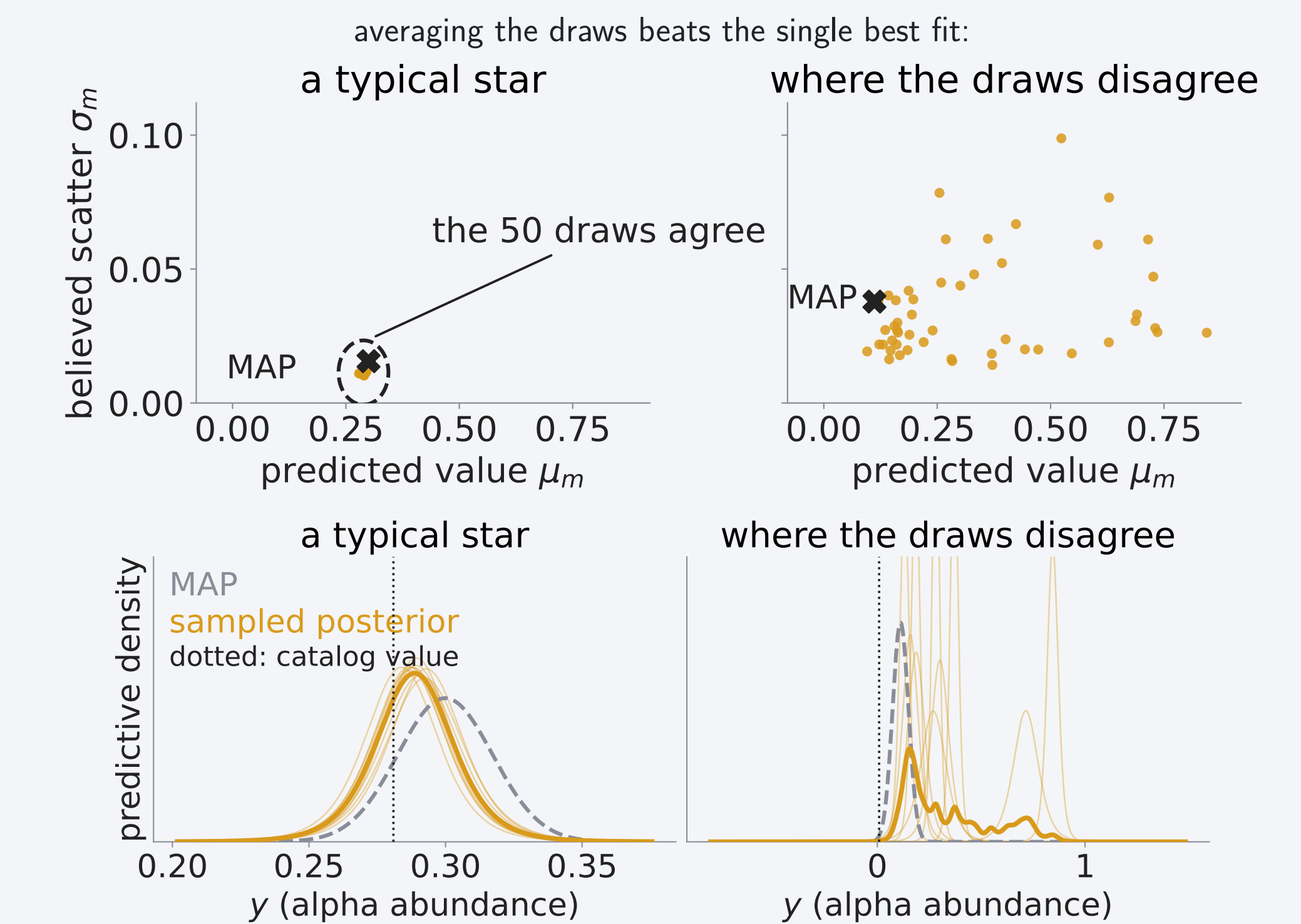


Empirical study on real Gaia data

We predict each star's alpha abundance plus a per-star scatter from its Gaia XP spectrum (110 coefficients), for 126,156 giants with APOGEE labels^[4,5], using a $D = 11,394$ network sampled with minibatch ray tracing.



$z \text{ std} = \text{actual} / \text{claimed error}$: closer to 1 is better.



Conclusion

- The ray moves at a fixed speed, so a noisy gradient can only change its direction. This is why the sampler is robust to noise.
- We took the ray tracing sampler from its paper^[2] to a real survey posterior. The draws are the product: averaging them beat the single best fit, and their spread warns us when the model is guessing.
- The gains held on real data: on average, held-out stars were 27% more probable under the posterior average than under the single best fit, most of it from the averaging alone.

References

- [1] Betancourt, M. 2017, *A Conceptual Introduction to Hamiltonian Monte Carlo*, arXiv:1701.02434
- [2] Behroozi, P. 2026, *The Ray Tracing Sampler: Bayesian Sampling of Neural Networks for Everyone*, Open Journal of Astrophysics 9, doi:10.33232/001c.165932
- [3] Chen, T., Fox, E. B., Guestrin, C. 2014, *Stochastic Gradient Hamiltonian Monte Carlo*, ICML, PMLR 32(2), 1683
- [4] Gaia Collaboration, Vallenari, A., et al. 2023, *Gaia Data Release 3: Summary of the content and survey properties*, A&A 674, A1
- [5] Laroche, A., Speagle, J. S. 2025, *Closing the Stellar Labels Gap: Stellar Label Independent Evidence for $[\alpha/M]$ Information in Gaia BP/RP Spectra*, ApJ 979, 5

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